

Who, not what: identity is recoverable but individual behavior is not in non-invasive and in-silico neuroforecasting

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Abstract

A growing market in “mind reading” and personalized neuro-mental-health reads an individual’s mental content or behavior from brain and biosignals, increasingly from a predicted, in-silico brain rather than a measured one. We argue that this crosses a boundary the data cannot support, and we make the boundary precise. Forecasting value decomposes into a population term and an individual term. The population term is tractable and, we prove, strictly improves with aggregation; the individual term requires between-subject variance that a zero-shot encoder does not have, so it is provably zero without per-subject enrollment. Three independent real datasets show the consequence: non-invasive readouts recover *who* a person is (in-scanner fingerprinting at 100%; a behavioral fingerprint at $14\times$ chance, $p < 0.001$) but not *what* they will do (the direct individual neural-to-behavior link abstains under a calibrated gate). A Bayes-optimal forecaster cannot escape this: given only population features it adds exactly zero individual value (recovery $r = 0.00$, incremental AUC = 0.00), and enrollment buys only a modest, saturating gain. Multimodal fusion does not rescue it, because added modalities inject nuisance variance faster than identifiable individual signal. These limits track the field’s own reliability results, under which individual brain-behavior effects are small and need thousands of subjects. We package the argument as a refusal-first read-out protocol and as an audit of personalized-neuro claims. The individual mental level is real and causally accessible, but today only to invasive methods, not to non-invasive or in-silico readouts.

1 Introduction

Products and clinical pipelines increasingly promise to read an individual’s emotion, intent, or future choice from brain signals, and to do so cheaply, often from a predicted brain: a stimulus-to-cortex encoder maps media or context to cortical activation, which is then read through region-to-construct labels. The promise is individual and mental. The question this paper answers is whether the individual mental level is reachable at all by such non-invasive, in-silico means.

Our answer is a boundary with a clean shape. Forecasting value splits into a population term and an individual

term, and the two are provably asymmetric. The population term is what classical neuroforecasting already exploits: aggregate neural signals predict markets and audiences even when the sampled individuals’ own behavior does not (Knutson et al., 2007; Genevsky et al., 2017; Dmochowski et al., 2014). The individual term is the one the market sells, and it is the one that is not there without a price. A zero-shot encoder emits the same features for every unseen person, so it carries no between-subject information, and the best-case individual forecast collapses to the population forecast. Paying the price, per-subject enrollment, recovers some individual signal, but on current encoders and signals it recovers identity far more than behavior.

Claims. (1) We state and machine-check the *individuation gap*: aggregation strictly improves the population term, while the zero-shot individual term is exactly zero (Sec. 2). (2) *Who, not what*: on real data, non-invasive readouts recover identity but not individual behavioral content (Sec. 3). (3) A Bayes-optimal forecaster adds zero individual value at zero-shot and only a modest, saturating gain with enrollment (Sec. 4). (4) Multimodal fusion does not rescue individuation, because variance accumulates faster than signal (Sec. 5). (5) These limits follow the field’s measurement-reliability results, and the gap is enforceable as a refusal-first protocol (Secs. 6–7).

2 The individuation gap

Model. Let an outcome be $Y = S + \varepsilon$ with idiosyncratic-noise variance $v > 0$, and let the predictor resolve an identifiable signal variance $a \geq 0$, the between-unit variance it can actually recover. The best-case squared correlation is

$$\rho^2(a, v) = \frac{a}{a + v}. \quad (1)$$

Aggregating over N units replaces v by v/N while keeping a ; a zero-shot encoder has $a = 0$.

Theorem 1 (Population term improves with aggregation). *For $a, v > 0$ and $N > 1$, $\rho^2(a, v) < \rho^2(a, v/N)$.*

Theorem 2 (Zero-shot individual term is null). *For all v, N , $\rho^2(0, v/N) = 0$.*

Theorem 3 (The individuation gap). *For $a, v > 0$ and $N > 0$, $\rho^2(0, v/N) < \rho^2(a, v/N)$.*

Theorem 2 is the impossibility: a predictor with no between-unit variance has zero best-case correlation with the individual component, for any noise level and any amount of pooling, because the quantity it would estimate is not present in the input. The three statements are formalized and machine-checked in Lean 4 with Mathlib (`IndividuationGap.lean`: `aggregation_improves`, `unseen_individual_is_zero`, `individuation_gap`). Figure 1b shows the gap against the measured individual-difference ceiling, which is itself small (Marek et al., 2022).

3 Who, not what

If the individual term is accessible only through enrollment, what does enrollment actually buy? Across three real paired datasets it buys identity, not behavior (Fig. 2).

Vision encoding (7T fMRI; NSD, BOLD5000). Subject-specific image-to-fMRI encoders ($N = 8$) encode well (median r 0.219 to 0.463, all positive). Cross-run fingerprinting identifies the subject in 25/25 regions at 100% (chance 1/8): enrolled identity is highly recoverable. The individual geometry that would carry behaviorally relevant content is reliably positive but small (+0.005 to +0.066 across regions), and the stimulus representation, not the loss, is the ceiling (Allen et al., 2022; Chang et al., 2019).

Value-based decision (fMRI and behavior; NARPS, ds000005). On 107 subjects and about 27,000 choices, a behavioral fingerprint identifies individuals at $14\times$ chance ($p = 5 \times 10^{-4}$): identity again recovers. Population links establish (held-out choice AUC 0.965, conformal coverage 1.00, ECE 0.034; loss aversion $\lambda = 1.45$ matching Tom et al., 2007; cross-lab transfer AUC 0.86 to 0.89; NAcc and anterior insula track loss, $p = 0.040$ and 0.025). The direct individual neural-to-behavior link abstains under a calibrated minimum-detectable-effect gate (Botvinik-Nezer et al., 2019). On the same gambles a content baseline dominates the aggregate, so a predicted index earns no neural credit.

The pattern is consistent: the *who* question (identity, verification) is answerable; the *what* question (this person’s mental content or next choice) abstains. Conflating the two is the central error of the personalized-neuro claim.

4 A Bayes-optimal forecaster cannot escape it

One might hope that a principled probabilistic model recovers the individual where point estimates fail. It does not, in the zero-shot regime, and the reason is information-theoretic rather than algorithmic. If the predicted feature carries no between-subject variance, then conditional on the population it is independent of the individual outcome, so the Bayes-optimal individual forecast equals the population forecast and the mutual information between feature and individual outcome is zero.

Proposition 1 (Zero-shot posterior collapse). *If the predictor is constant across subjects, the posterior over an individual’s outcome equals the population prior, and no estimator achieves above-population individual accuracy.*

We verify this with a Bayesian simulation (sigmoidal choice with a latent per-subject bias; a grid posterior updated from a subject’s own trials). At zero-shot the recovery correlation is $r = 0.00$ and the incremental AUC over the population model is +0.000. Enrollment pays the price: recovery rises to $r = 0.58, 0.86, 0.99$ at 2, 10, 200 trials, and the incremental AUC rises to +0.05, +0.11, +0.14 and saturates (Fig. 3). The honest reading is twofold. Zero-shot individual probabilistic prediction is falsified, exactly. Enrolled individual prediction is possible but modest and data-hungry, and it saturates well short of certainty.

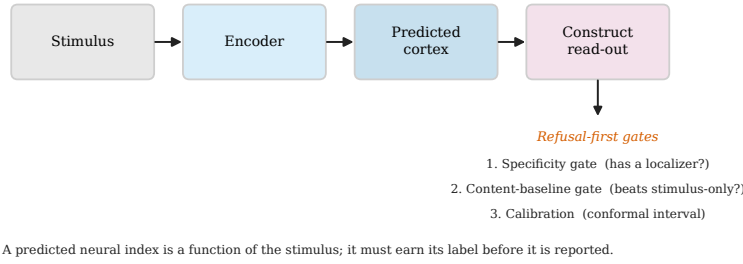
5 Does multimodal fusion rescue it?

A natural rescue is to add EEG, heart rate, electrodermal activity, or hormonal assays. Under independence, inverse-variance fusion lowers variance: the fused variance is at most that of any single modality, a fact we machine-check for the general finite case (`FusionMath.lean`, `fused_le_min`). The rescue fails in practice because the independence premise does not hold and because each added modality injects its own nuisance variance.

Proposition 2 (Fusion is not monotone in the number of modalities). *With fused signal $A = \sum_m a_m$ and fused effective noise $v + \sum_m \delta_m$, where δ_m is the nuisance variance the m th modality injects, the individual correlation $\rho^2 = A/(A + v + \sum_m \delta_m)$ increases when modality m is added if and only if a_m/δ_m exceeds the current signal-to-noise ratio; low-reliability modalities strictly lower it.*

Figure 4 shows the consequence: adding high-reliability modalities helps, but low-reliability ones (single-channel autonomic signals, slow and state-dependent hormonal

a In-silico neuroforecasting pipeline



b The individuation gap

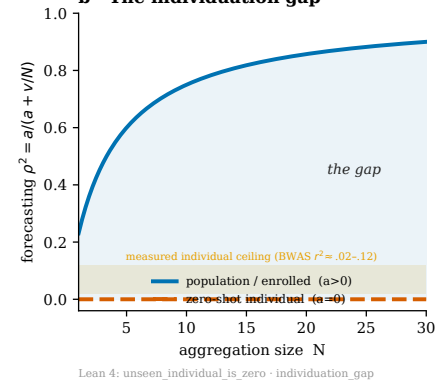
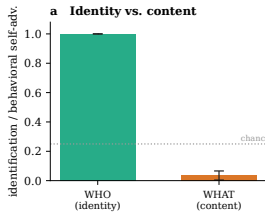
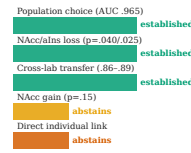


Figure 1: (a) The in-silico pipeline and the refusal-first gates. (b) The individuation gap: the population term rises with aggregation while the zero-shot individual term is pinned at zero, below the small measured individual-difference ceiling (Theorems 1–3).

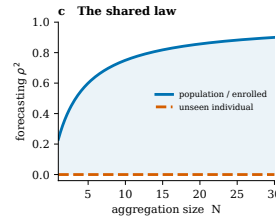
a Identity vs. content



b Value-based decision



c The shared law



a Enrolled identity is recoverable (fingerprinting 25/25 ROIs, 100%; behavioral fingerprint 14× chance, $p < .001$); individual behavioral content is not. b Population & cross-lab links establish; the direct individual neural–behavior link abstains under the calibrated gate.

Figure 2: Who, not what. (a) Identity is recoverable (in-scanner fingerprint 100%; behavioral fingerprint 14× chance); individual behavioral content is not. (b) Population and cross-lab links establish; the direct individual link abstains. (c) Both instantiate the proven gap.

Bayes-optimal individual prediction: the price of enrollment

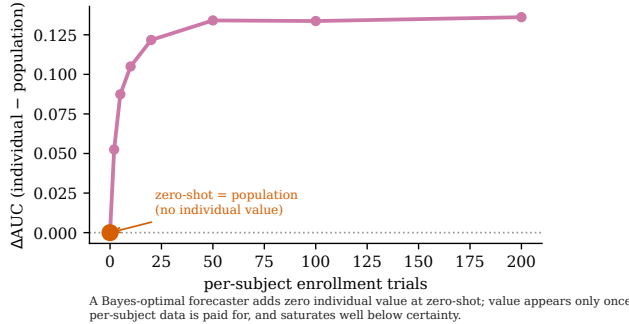


Figure 3: A Bayes-optimal individual forecaster adds exactly zero value at zero-shot and only a modest, saturating gain once per-subject data is paid for.

Multimodal fusion is not a free lunch

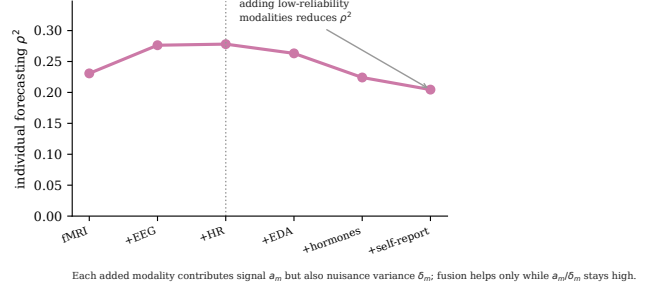


Figure 4: Multimodal fusion is not a free lunch. Each added modality contributes signal a_m but also nuisance variance δ_m ; low-reliability modalities lower individual ρ^2 .

6 Why the individual level is hard

The gap is not an artifact of one pipeline. The field’s measurement literature predicts it. Brain-wide association studies have small effect sizes (univariate $r \approx 0.14$, multivariate $r \approx 0.34$) and need thousands of subjects for reproducibility, while within-person and interventional effects are larger (Marek et al., 2022; Kang et al., 2024). Common task-fMRI individual measures have poor test-

assays) push ρ^2 back down. Fusion can lift the population term; it does not supply the between-subject signal the individual term needs, and past a point it hurts.

World / regime	Gate 1	Gate 2	Expected
SIGNAL/enrolled	pass	pass +0.449	fire
SIGNAL/unseen	no	no +0.028	silent
NULL/enrolled	pass +0.087	no -0.001	silent
NULL/unseen	pass +0.034	no -0.001	silent

Table 1: Gate 2 fires only where individual signal exists and the subject is enrolled (CI [+0.378, +0.526]). Gate 1 passes even in NULL, so it measures population value and Gate 2 measures individual value.

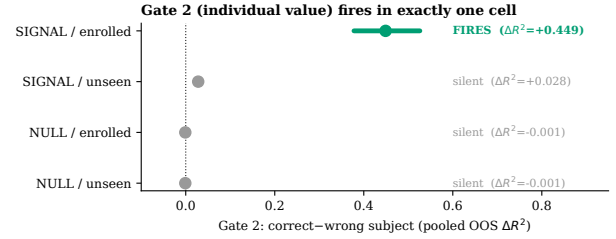
retest reliability (mean ICC 0.397), which bounds any individual-differences use (Elliott et al., 2020), though multivariate signatures can do better (Kragel et al., 2021). Individual precision functional mapping is achievable but requires hours of per-subject data (Gordon et al., 2017; Du et al., 2025). All three say the same thing as Eq. 1: the individual term is small and expensive, and a zero-shot predictor pays none of the price.

7 The refusal-first protocol

The practical upshot is a read-out that withholds by default (Fig. 1a). A construct score is returned only if it passes a specificity gate (the construct term has a meta-analytic localizer and its named regions are the term’s best match), a content-baseline gate (the predicted index beats a stimulus-only model out of sample), and a calibration requirement (a split-conformal interval; Vovk et al., 2005). Separately, the protocol reports identity and content on different channels, so a system may verify *who* without claiming to read *what*. The specificity gate alone withholds, for example, a manipulation index, because no localizer for persuasion exists.

8 Synthetic validation of the apparatus

We verify the control ladder recovers known truth before trusting it. In a crossed design (WORLD: signal vs. null; SPLIT: enrolled vs. unseen), with ridge regression, pooled out-of-sample R^2 , and a paired bootstrap over 20 seeds, Gate 2 (individual value, correct minus wrong subject) fires in exactly the one cell where individual signal exists and the subject is enrolled, with no false positives (Table 1, Fig. 5). In the unseen regime Gate 2 cannot fire, which is Theorem 2 in simulation. These numbers are synthetic and validate the apparatus, not any brain.



20 seeds, paired bootstrap on synthetic ground truth. Zero false positives in the NULL worlds; silent in SIGNAL/unseen because a zero-shot encoder cannot individuate an unseen subject.

Figure 5: Synthetic apparatus check. Gate 2 (individual value) fires only in SIGNAL/enrolled; it is silent in both NULL cells and in SIGNAL/unseen.

9 What would reach the individual level

The individual mental level is real and causally accessible, but today to invasive methods. Optogenetics achieves cell-type-specific causal control of circuits and behavior, with brain-wide readout when paired with fMRI, in animals (Zhao et al., 2011; Bernal-Casas et al., 2017). Intracortical brain-computer interfaces decode attempted speech and movement from a single person’s spiking activity at high accuracy and in daily home use (Willett et al., 2023; Card et al., 2025; Silva et al., 2024). These reach the individual precisely because they record or perturb that individual’s own circuitry, which is the between-subject signal a zero-shot non-invasive readout lacks. They mark the direction in which small individual signals may eventually be tracked, and they are out of scope for the non-invasive consumer and clinical claims this paper audits.

10 Related work

Measured-brain neuroforecasting (Knutson et al., 2007; Dmochowski et al., 2014; Yao et al., 2024), reverse-inference caution (Poldrack, 2006; Hutzler, 2013), conformal prediction (Vovk et al., 2005), reliability and sample-size results (Marek et al., 2022; Elliott et al., 2020; Gordon et al., 2017), and inverse-variance fusion are prior work. The field treats population versus individual as a design trade-off. We argue that for non-invasive and in-silico readouts it is a boundary with a provable shape, separate for identity and content, unrescued by fusion, and enforceable as a protocol.

11 Limitations

The synthetic ladder and the Bayesian simulation validate the apparatus and the probability claim, not any specific product. The real-data arms were analyzed under separate protocols in prior work, so they are convergent evidence rather than one preregistered study; the

decision-fMRI individual link abstains at the boundary of power, which is an abstention, not a positive null. The fusion proposition is elementary and not machine-checked beyond the independence lemma. Reported numbers were read from the analysis repositories and should be regenerated from frozen commits before camera-ready.

12 Conclusion

Non-invasive and in-silico brain readouts can tell you who someone is and can forecast a crowd. They cannot tell you what an individual is thinking or will choose, because the individual signal is not in the input without a per-subject price, and that price currently buys identity rather than behavior. Fusion does not change this, a Bayes-optimal forecaster does not escape it, and the measurement literature already said so. A read-out that respects the boundary returns fewer numbers, and the ones it returns have earned their label.

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